## ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** zero | cos 90 = 0. |
| 2 | **(a)** 9 | 0.06 x 150. |
| 3 | **(d)** 4 | Effort arm/load arm. |
| 4 | **(c)** 20 m/s | sqrt(2gh). |
| 5 | **(b)** 120 J | Only vertical displacement counts. |
| 6 | **(a)** 45 rupees | 9 units x 5. |
| 7 | **(c)** 50 N | MA = 4. |
| 8 | **(c)** 36 per cent | (5 - 3.2)/5. |
| 9 | **(d)** 5 m/s | v = sqrt(2KE/m). |
| 10 | **(a)** 500 W | 15000/30. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | W = 200 J (1); KE = 200 J (1). | 1+1 |
| 12 | KE = 450 J (1); F = 450/0.03 = 15000 N (1). | 1+1 |
| 13 | MA = 4 (1); Effort = 100 N (1). | 1+1 |
| 14 | W = 500 J (1); P = 100 W (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Start: KE 225 J (1); At 5 m: PE 100 J, KE 125 J (1); Top: PE 225 J (11.25 m) (1). | 1+1+1 |
| 16 | At 50 cm: 40 N, MA 2.5 (1); At 1 m: 20 N, MA 5 (1); Longer effort arm gives larger MA (1). | 1+1+1 |
| 17 | KE is lost on impact (1); Converts to heat, sound, deformation (1); Energy is conserved overall (1). | 1+1+1 |
|  | **OR:** W = 100000 J (1); P = 2500 W (1); mgh (1). | 1+1+1 |
| 18 | Work in = work out (1); E x 5 = 300 x 1 (1); E = 60 N; MA = 5 (1). | 1+1+1 |
|  | **OR:** PE lost = 600 J (1); KE gained = 360 J (1); Work against friction = 240 J (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Start: PE 900 J (1); After 1 s: fallen 5 m, KE 100 J, PE 800 J (1); After 2 s: fallen 20 m, KE 400 J, PE 500 J (1); Ground: KE 900 J (v = 30 m/s) (1); Total 900 J (1). | 1+1+1+1+1 |
|  | **OR:** Efficiency = useful output / input x 100 (1); = 600/800 x 100 (1); = 75 per cent (1); Energy lost = 800 - 600 (1); = 200 J, wasted mainly as heat by friction (1). | 1+1+1+1+1 |
| 20 | PE of water to KE to electricity (1); Power = 5000 x 10 x 40 = 2 x 10^6 W (1); Useful power = 1.6 x 10^6 W (1); Efficiency 80 per cent (1); Renewable (1). | 1+1+1+1+1 |
|  | **OR:** Work A = 240000 J, P = 400 W (1); Work B = 200000 J, P = 555.6 W (1); B is more powerful (1); Ratio A:B = 0.72 (1); P = mgh/t (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of $3.6 \times 10^6$ J, or calling kWh a unit of power.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 3 | L/E. |
| 2 | **(a)** 2 m/s | sqrt(2 x 10 x 0.2). |
| 3 | **(c)** four times | KE proportional to v squared. |
| 4 | **(b)** 200 W | W/t. |
| 5 | **(a)** 4 | l/h. |
| 6 | **(b)** 45 m | u squared/2g. |
| 7 | **(b)** half | 2 x 1/4. |
| 8 | **(c)** 0.3 | 0.06 kW x 5 h. |
| 9 | **(c)** 80 per cent | 400/500 x 100. |
| 10 | **(d)** elastic potential energy | Stored. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Larger KE (1); Work = F s = KE, so s larger (1). | 1+1 |
| 12 | Friction and weight of moving parts waste energy (1); Output work is less than input (1). | 1+1 |
| 13 | v = sqrt(2 x 10 x 5) (1); = 10 m/s (1). | 1+1 |
| 14 | Gain in PE is the same (1); Work in = mgh in both cases (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | PE = KE (1); v = 3 m/s (1); PE to KE (1). | 1+1+1 |
| 16 | v = 10 m/s (1); KE = 150 J (1); Fall in 1 s: 5 m, PE lost 150 J (1). | 1+1+1 |
| 17 | Some KE is lost to heat and sound (1); Less energy for the next rise (1); Total energy is conserved (1). | 1+1+1 |
| | **OR:** KE1 = 1000 J (1); KE2 = 4000 J (1); Work = 3000 J (1). | 1+1+1 |
| 18 | Friction: negative work (1); Centripetal force perpendicular to displacement: zero work (1); Moon in circular orbit: zero (1). | 1+1+1 |
| | **OR:** 12 J (1); 12 J (1); Total 24 J (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | W by man = 1500 J (1); Against gravity = 1000 J (1); Against friction = 500 J (1); Efficiency = 66.7 per cent (1); MA = 500/150 = 3.3 (ideal 5) (1). | 1+1+1+1+1 |
| | **OR:** mgh = (1/2) m v squared (1); v = sqrt(2gh) (1); h = 5: 10 m/s (1); h = 20: 20 m/s (1); Independent of mass and shape (1). | 1+1+1+1+1 |
| 20 | Moment of load = moment of effort (1); MA = effort arm/load arm (1); Effort arm = 1.3 m (1); Effort = 600 x 0.2/1.3 = 92.3 N (1); MA = 6.5 (1). | 1+1+1+1+1 |
| | **OR:** KE = 625 J (1); F = 6250 N (1); t = 0.0008 s (1); At 500 m/s: KE 2500 J, F 25000 N (4 times) (1); Distance four times if the same force (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Saying work is done by holding a load or by pushing a wall; without displacement, work is zero.
- Treating work done by friction or by a force perpendicular to the motion (a coolie walking with a load on his head) as positive.
- Writing $KE = mv^2$ or taking $KE \propto v$; doubling the speed makes the kinetic energy four times.
- Not converting units: minutes to seconds in $P = W/t$, kW to W, or 20 cm of stairs to metres.
- Believing that a machine (lever, ramp) saves work; it only reduces the effort by increasing the distance moved by the effort.
- Using $1\ \text{kWh} = 1000$ J instead of $3.6 \times 10^6$ J, or calling kWh a unit of power.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Work, Energy and Simple Machines** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 0 | v = 0. |
| 2 | **(b)** 14.4 x 10^6 J | 2000 x 7200. |
| 3 | **(c)** 80 per cent | Output 200 J, input 250 J. |
| 4 | **(c)** 500000 J | mgh. |
| 5 | **(a)** 80 J | mg x 4. |
| 6 | **(d)** twice the power of B | Same work, half the time. |
| 7 | **(a)** 100 per cent | No losses. |
| 8 | **(a)** 17.3 m/s | sqrt(2 x 10 x 15). |
| 9 | **(a)** 8.9 m/s | sqrt(2 x 10 x 4). |
| 10 | **(b)** 5000 W | P = F v. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Output 800 J, input 1000 J (1); Efficiency 80 per cent (1). | 1+1 |
| 12 | 0.5 kWh (1); 1.8 x 10^6 J (1). | 1+1 |
| 13 | By man 100 J (1); By friction -40 J (1). | 1+1 |
| 14 | Longer effort arm (1); Less effort needed (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Saved power 32 W (1); 32 x 5 x 30 = 4.8 kWh (1); Saving = 28.8 rupees (1). | 1+1+1 |
| 16 | W = mgh = 60 x 10 x 100 = 60000 J (1); t = 600 s (1); P = 60000/600 = 100 W (1). | 1+1+1 |
| 17 | Moments equal (1); 30 x 2 = 40 x d (1); d = 1.5 m (1). | 1+1+1 |
|  | **OR:** Hammer has KE at the moment of impact (1); KE gives large force over short distance (1); Work done on the nail = F x d (1). | 1+1+1 |
| 18 | E = 2 x 0.5 x 30 = 30 kWh (1); Cost = 180 rupees (1); Units = kWh (1). | 1+1+1 |
|  | **OR:** MA = R/r = 4 (1); Effort = 50 N (1); Ideal machine (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | At the top PE max (1); At the valley v = sqrt(600) = 24.5 m/s (1); At the 20 m hill: v = sqrt(2 x 10 x 10) = 14.1 m/s (1); KE and PE exchange (1); Friction ignored (1). | 1+1+1+1+1 |
|  | **OR:** At the top: gravitational PE max (1); Falling: PE to KE (1); Trampoline: KE to elastic PE (1); Then back to KE and PE (1); Some energy lost as heat/sound (1). | 1+1+1+1+1 |
| 20 | F x 10 = 5 x 10 x 4 (1); F = 20 N (1); W = 200 J (1); PE = 200 J (1); MA = 2.5 (1). | 1+1+1+1+1 |
|  | **OR:** Work = 100000 J (1); Human: 1000 s (1); Motor: 100 s (1); Engine: 1 s (1); Time = W/P (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

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